

Shreyas Kalvankar

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Education

Technische Universiteit Delft MSc Computer Science: Machine Learning Applications Engineering & Algorithmics Master's thesis: Neural network training dynamics and spectral bias Investigated spectral bias and training dynamics of overparameterized neural networks, combining NTK theory, Fourier/operator analysis, and numerical experiments to study how finite width and sampling affect feature learning.	September 2024 – July 2026 Delft, Netherlands
Savitribai Phule Pune University Bachelor of Engineering in Computer Engineering Bachelor's thesis: Generative Computer Image Colorization and Super-resolution using GANs	August 2017 – May 2021 Pune, India

Experience

Technische Universiteit Delft Teaching Assistant Supported MSc courses in Machine and Deep Learning, Statistical Learning, Security and Cryptography, and Cybersecurity. Solving. • Led tutorials, created and graded assignments and exams, and assisted students with debugging implementations.	September 2025 – Present Delft, Netherlands
Dalton Maag Ltd. Software Developer Automated OCR (Chinese-Japanese-Korean) glyph generation using a Python genetic-algorithm POC, demonstrating reduction of production time from months to minutes. • Developed core features for PriceBot (TypeScript, Ruby on Rails) and productionized it, cutting quote generation time from hours to seconds. • Designed a containerized GitLab CI/CD pricing health monitor with daily metric scraping and threshold alerting via emails that prevented silent regressions and potential six-figure revenue losses. • Engineered a Devanagari complexity model and overhauled Arabic, Greek, and Cyrillic models using TypeScript and Ruby on Rails/PostgreSQL, scaling pricing coverage from about 300 to about 2,500 glyphs. • Modeled effort-scaling heuristics and non-linear design constraints as a dependency graph to automate task scheduling, replacing a fully manual estimation process and reducing planning time from hours to minutes. • Contributed to DSedit, an internal ElectronJS, VueJS, and Django platform, abstracting complex Unified Font Object (UFO) and Designspace specifications into a reliable GUI to eliminate manual XML editing and compilation errors.	November 2021 – February 2024 London, United Kingdom
Relfor Labs Pvt. Ltd. Machine Learning Research Scientist (Consulting, Part-time) Designed compact, decomposable CNN architectures in PyTorch for edge-oriented deployments, enabling low-latency audio transition detection by caching hierarchical time-series embeddings during rolling-window inference. • Built PyTorch Lightning and Distributed Data Parallel training pipelines on NVIDIA DGX A100, accelerating experimentation by about 50%. • Accelerated terabyte-scale (13TB) data preprocessing by 8x using Numba JIT compilation and Pandas to optimize CPU utilization. • Reduced pipeline storage footprint by 30% by migrating UTF-8 data to Apache Arrow, implementing byte-addressable binary storage for rapid memory loading.	September 2022 – Present Pune, India
Relfor Labs Pvt. Ltd. Machine Learning Engineer Designed CNN architectures for audio classification using mel-spectrograms in PyTorch, steering experimental design to surpass standard baselines, achieving about 98.6% accuracy and about 0.98 F1-score.	August 2021 – November 2021 Pune, India

Technical Skills

Languages: Python, JavaScript, TypeScript, C, C++, Rust, Go
Machine Learning: PyTorch, TensorFlow, Keras, JAX, Optuna
Backend and Web: Ruby on Rails, React, Vue, Django, HTML, CSS
Data and Databases: PostgreSQL, MongoDB, Pandas, Apache Arrow
DevOps and Infrastructure: Docker, CI/CD, container orchestration, observability
Cloud and Compute: Google Cloud Platform, NVIDIA DGX A100, Distributed Data Parallel

Projects

BingeBuddy Built BingeBuddy, a task-oriented conversational recommendation system for personalized movie and TV series discovery. Github: github.com/kariga8861/binge-buddy
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- Designed the agent memory architecture to track and recall relevant information from past conversations, using LLM-based workflows to decide what should be stored in long-term memory, what to extract, and how to categorize and merge new information with existing memories.
- Developed the Django backend and used LangChain to orchestrate the agentic workflows for recommendation and memory management.
- Used MongoDB as the long-term memory store and ReactJS for the frontend, enabling more personalized multi-turn recommendations based on prior interactions.

REMLA Project

Open Source github.com/remla25-team15/openremla

- Built and deployed a production-oriented machine learning application from scratch with a focus on reproducibility and scalability.
- Used Vagrant, Ansible, Kubernetes, Helm, Istio, DVC, and Git-based CI/CD to manage infrastructure, versioning, and deployment workflows.
- Developed experience with release engineering practices for machine learning systems in a team-based software setting.

Galaxy Morphology Classification

Research Project github.com/shinichi/GalaxyEfficientNet

- Built galaxy morphology classification, based on Kaggle Galaxy Zoo 2 competition, implementing the EfficientNet architectures in Tensorflow.
- Built TensorFlow CNNs for galaxy morphology prediction, achieving a top-3 Kaggle ranking for vote-fraction regression with RMSE 0.07765.
- Developed a 7-class galaxy morphology classifier with 93.7% accuracy and 0.8857 F1-score.

EinsteinPy

Open Source github.com/institupr/einsteinpy

- Contributed core general relativity metrics and numerical fixes to a community Python library used for scientific computing and education.
- Added the Reissner–Nordström metric, a static solution to the Einstein–Maxwell field equations.
- Fixed issues in the Kerr–Newman and Kerr metric classes.
- Implemented calculations for the event horizon and ergosphere of a Kerr–Newman black hole.

Publications & Pre-prints

- **Astronomical Image Colorization and Up-scaling with Conditional Generative Adversarial Networks**, INFOR-MATIK 2022
- **EinsteinPy: A Community Python Package for General Relativity**, arXiv:2005.11288
- **Galaxy Morphology Classification using EfficientNet Architectures**, arXiv:2008.13611

Languages

English (Fluent, IELTS 8.5, CEFR C2), Dutch (Elementary), Japanese (Elementary)